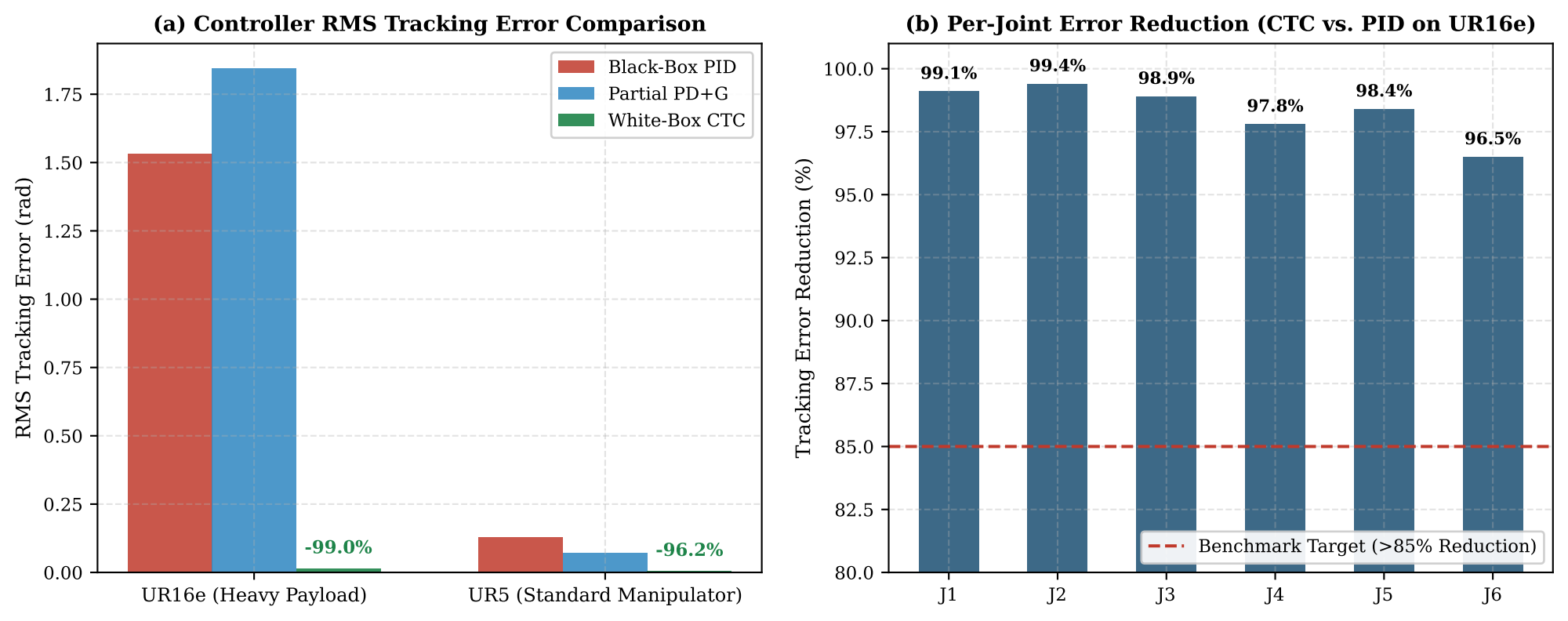
(a) Controller RMS Tracking Error Comparison
RMS Tracking Error (rad)
Black-Box PID
Partial PD+G
White-Box CTC
-99.0%
-96.2%
UR16e (Heavy Payload)
UR5 (Standard Manipulator)
(b) Per-Joint Error Reduction (CTC vs. PID on UR16e)
Tracking Error Reduction (%)
99.1%
99.4%
98.9%
97.8%
98.4%
96.5%
Benchmark Target (>85% Reduction)
J1
J2
J3
J4
J5
J6